

#This code will be used to demonstrate theorem 6.20. We will demonstrate that $m1 = ((z + w)^2 - y^2) \cdot ((w + z)^3 - y^3)^2 \cdot ((x^3 - z^3) \cdot w^2 + (x^3 - w^3) \cdot z^2)^2 / (z \cdot w \cdot (w + z)^4 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))^2)$ is increasing along the $\mathcal{V}_2$ curve; (Part 2 – discarding potential critical points)

#Define r2

$$\begin{aligned}
 r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\
 & (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\
 & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\
 & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\
 & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\
 & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\
 & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\
 & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\
 & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\
 & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
 & - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\
 & - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
 & - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
 & + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
 & - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
 & - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
 & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
 & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
 & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6
 \end{aligned}$$

$$\begin{aligned}
& + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\
& \left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big):
\end{aligned}$$

#The candidate points (a, b) for the intersection of $r_2 = 0$ and $\tilde{f} = 0$ are of the form $a \in A$ and $b \in B$ where

$A := [3.474957787, 1.024339148, 1.486779477, 2.120845040, 2.648207398, 3.867159438] :$

$B := [1.338933212, 1.386399815, 1.398181897] :$

$Points := [seq(seq([a, b], b \text{ in } B), a \text{ in } A)] :$

$Points_filtered := select(p \rightarrow p[2] < p[1], Points);$

$Points_filtered := [[3.474957787, 1.338933212], [3.474957787, 1.386399815], [3.474957787, 1.398181897], [1.486779477, 1.338933212], [1.486779477, 1.386399815], [1.486779477, 1.398181897], [2.120845040, 1.338933212], [2.120845040, 1.386399815], [2.120845040, 1.398181897], [2.648207398, 1.338933212], [2.648207398, 1.386399815], [2.648207398, 1.398181897], [3.867159438, 1.338933212], [3.867159438, 1.386399815], [3.867159438, 1.398181897]]$ (1)

with(plots) :

with(plottools) :

$r := a - b :$

$p1 := \text{implicitplot}([r_2 = 0, r = 0],$
 $a = 1 \dots 4, b = 1 \dots \text{sqrt}(3),$
 $\text{scaling} = \text{constrained},$
 $\text{gridrefine} = 2,$
 $\text{thickness} = 2,$
 $\text{color} = [\text{blue}, \text{red}],$
 $\text{labels} = ["a", "b"],$
 $\text{title} = \text{"Graphic for } r_2 = 0 \text{"}$
 $) :$

rectangles

$R1 := \text{rectangle}([1, 1], [1.5, 1.5],$
 $\text{color} = \text{silver}, \text{transparency} = 0.5,$
 $\text{linestyle} = 3, \text{ # pontilhado}$
 $\text{thickness} = 2) :$

```

R2 := rectangle( [2.5, 1.3], [4, 1.5],
    color = silver, transparency = 0.5,
    linestyle = 3, # pontilhado
    thickness = 2 ) :

```

points

```

P := pointplot(
    Points_filtered,
    symbol = solidcircle,
    symbolsize = 12,
    color = black
) :

```

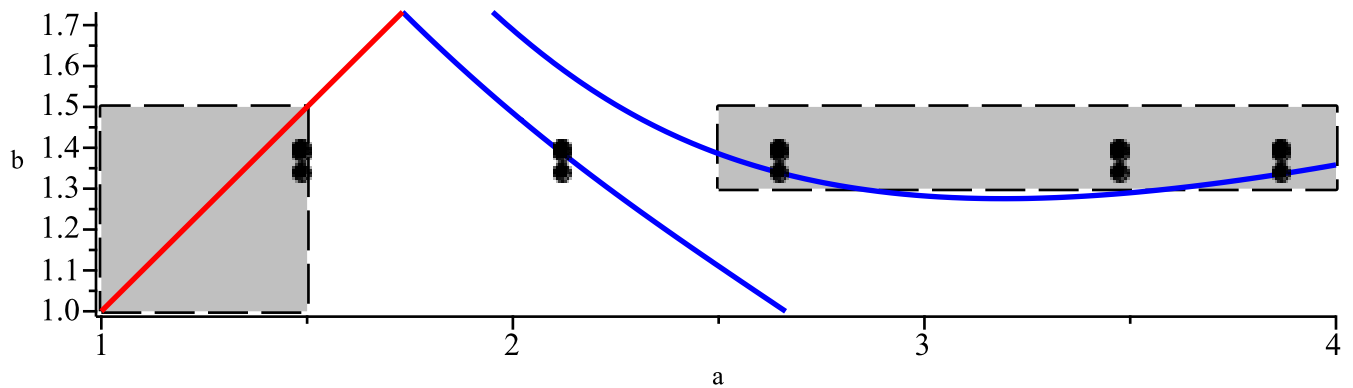
sobrepor tudo

```

display( [p1, P, R1, R2] );

```

Graphic for $r_2=0$



```

Points_filtered2 := [[2.120845040, 1.338933212], [2.120845040, 1.386399815], [2.120845040,
    1.398181897] ] :

```

```

p2 := implicitplot(
  [r2=0, r=0],
  a=2.10 .. 2.15, b=1.3 .. 1.42,
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, red],
  labels=["a", "b"],
  title="Graphic for r2=0"
) :

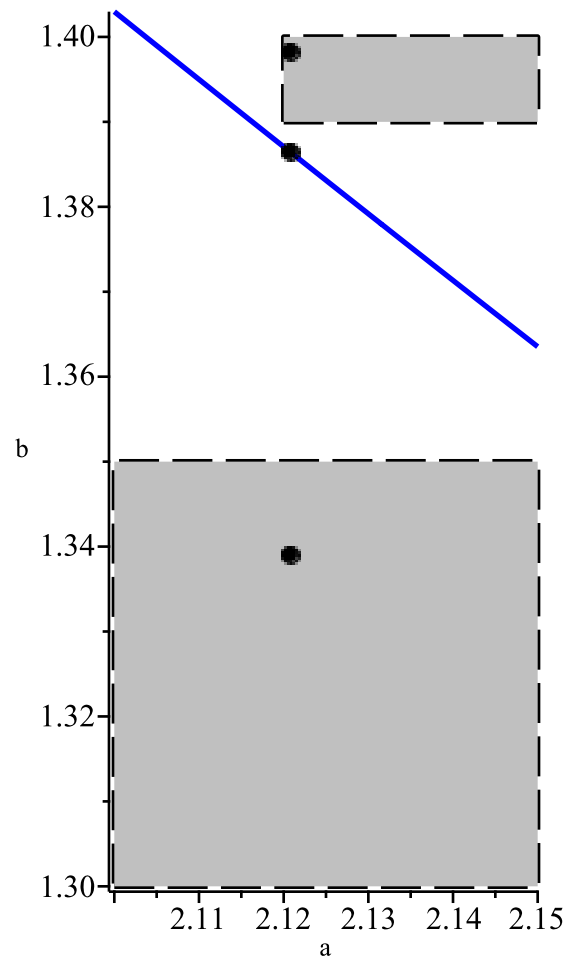
# rectangles
R3 := rectangle([2.10, 1.3], [2.15, 1.35],
  color=silver, transparency=0.5,
  linestyle=3, # pontilhado
  thickness=2) :
R4 := rectangle([2.12, 1.39], [2.15, 1.4],
  color=silver, transparency=0.5,
  linestyle=3, # pontilhado
  thickness=2) :

# points
P2 := pointplot(
  Points_filtered2,
  symbol=solidcircle,
  symbolsize=12,
  color=black
) :

# sobrepor tudo
display([p2, P2, R3, R4]);

```

Graphic for $r_2=0$



#By covering the previous points with rectangles, we will show that the curve $r_2=0$ does not intersect them.

restart;

#Define r2

$$\begin{aligned}
 r_2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\
 & -4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\
 & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\
 & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\
 & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\
 & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\
 & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\
 & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\
 & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\
 & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
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 & - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
 & + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
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 & - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
 & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
 & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
 & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6
 \end{aligned}$$

$$\begin{aligned} &+ 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\ &+ 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\ &\left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big): \end{aligned}$$

#Apply the Möbius transformation on the rectangle $(1, 1.5) \times (1, 1.5)$ that contains three points, constructing r_2

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{1 + \frac{3}{2} \cdot c}{1 + c}, \\ b = \frac{1 + \frac{3}{2} \cdot d}{1 + d} \end{array} \right\} :$$

#Compute the degree of r^2 in c

$$degree(r2, c);$$

32

(2)

#Define the coefficients $c_i = \text{coeff}(r^2, c, i)$

for i from 0 to 32 do
$$c[i] := \text{coeff}(\text{`r2`}, c, i);$$

od:

#Compute the all degree of c_i in d

```
seq( degree( c[i], d ), i = 0 .. 32 );
```

[illegible]

#From here we will analyze the signal variation of all coefficients c_i

[illegible]

[illegible]

[illegible]

#By covering the previous points with rectangles, we will show that the curve $r_2=0$ does not intersect them.

restart;

#Define r2

$$\begin{aligned}
 r_2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\
 & -4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\
 & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\
 & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\
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 & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
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 & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
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 \end{aligned}$$

$$\begin{aligned} &+ 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\ &+ 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \\ &\left. + \frac{8}{5} b^2 - 1 \right) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big): \end{aligned}$$

#Apply the Möbius transformation on the rectangle $(2.1, 2.15) \times (1.3, 1.35)$ that contains three points, constructing r_2

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{\frac{21}{10} + \frac{43}{20} \cdot c}{1 + c}, \\ b = \frac{\frac{13}{10} + \frac{27}{20} \cdot d}{1 + d} \end{array} \right\} :$$

#Compute the degree of r^2 in c

$$degree(r2, c);$$

32

(5)

#Define the coefficients $c_i = \text{coeff}(r^2, c, i)$

for i from 0 to 32 do
$$c[i] := \text{coeff}(\text{'r2'}, c, i);$$

od:

#Compute the all degree of ci in d

```
seq( degree( c[i], d ), i = 0 .. 32 );
```

[illegible]

#From here we will analyze the signal variation of all coefficients c_i

[illegible]

[illegible]

[illegible]

#By covering the previous points with rectangles, we will show that the curve $r_2=0$ does not intersect them.

restart;

#Define r2

$$\begin{aligned}
 r_2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\
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 & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\
 & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\
 & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
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 & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
 & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\
 & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6
 \end{aligned}$$

[illegible]

[illegible]